Latent Predictive Modeling vs. High-Fidelity Visual Reconstruction for Robust Surgical Phase Semantics

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Project Definition

Automated classification of surgical workflows (CAI) is vital for real-time endoscopic video analysis [6]. This study compares two paradigms:

- Pixel-based Reconstruction: Learning by recreating every frame.
- Latent-Predictive Modeling: Learning by predicting future states in an abstract embedding space.

Motivation

Societal: In the domain of CAI, the robust recognition of surgical workflow is not merely a technical challenge but a fundamental requirement for modernizing the operating room [3]. Automation can reduce preventable errors, optimize OR scheduling, and provide formative feedback via Video-Based Assessment (VBA).

Technical: The technical motivation stems from limitations in current AI models. Current models (like GSViT) waste capacity on "noise" (smoke, blood, reflections) due to their learning objective. Surgical videos are frequently degraded by noise such as smoke or blood splashes [10]. This study hypothesizes that modeling latent motion is more effective than high-fidelity reconstruction for capturing the "universal flow" of surgery. Consequently, the project proposes a shift toward latent-predictive models, specifically V-JEPA [2]. Furthermore, to understand representation differences between the two learning objectives, dimensionality reduction and layer-wise probing [7] will be utilized.

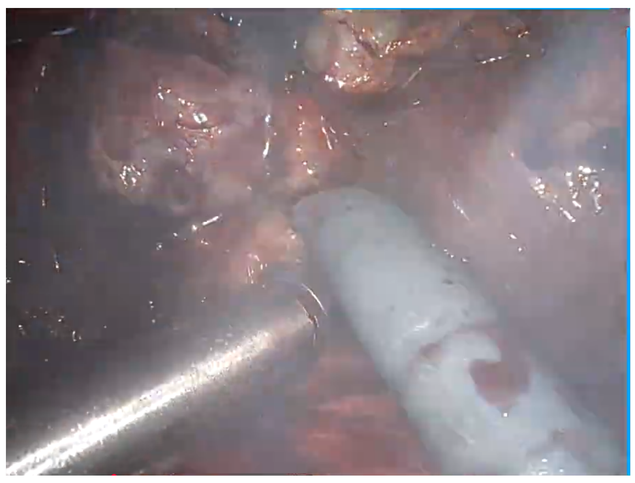

Figure 1. Smoke during Surgery

Research Questions

To what extent does a latent-predictive objective improve the semantic robustness and topological organization of surgical phase representations compared to traditional pixel-reconstruction approaches?

1. To what extent does the choice of pre-training objective, latent future prediction versus pixel reconstruction, impact the performance of learned representation for down-stream task?
2. To what extent does the choice of pre-training objective affect the clustering quality of surgical phase embeddings?

Comparison of Architectures

GSViT (Pixel-Based Baseline)

- Uses EfficientNet ViT for speed (10k+ FPS).
- Pretrained on next-frame reconstruction with 1-frame input and 1-frame output.
- Benchmarked on Cholec80 [8] dataset, and achieved 86.3% accuracy.
- Weakness: Prioritizes visual texture; sensitive to surgical noise (smoke/glare).

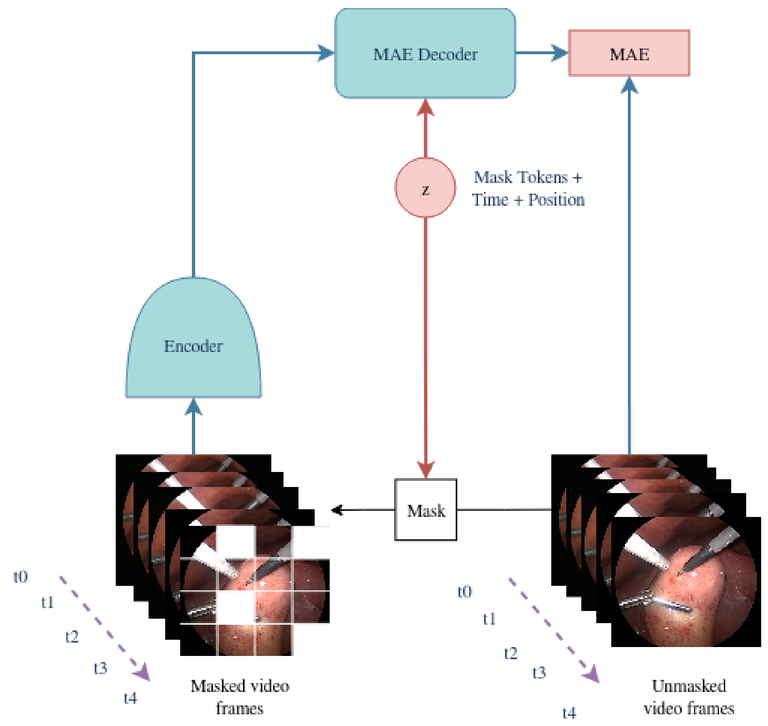

Figure 2. GSViT Architecture

V-JEPA (Latent-Based Approach)

- Argument stems from the idea that human-like intellgience emerges from unsupervised world models.
- JEPA focuses on predicting missing information within the latent-space
- V-JEPA specifically adapts this architecture for the temporal domain using ViT (Vision Transformer)
- Strength: Acts as a "filter" for noise, focusing on semantic instrument-tissue interactions. More computationally efficient.

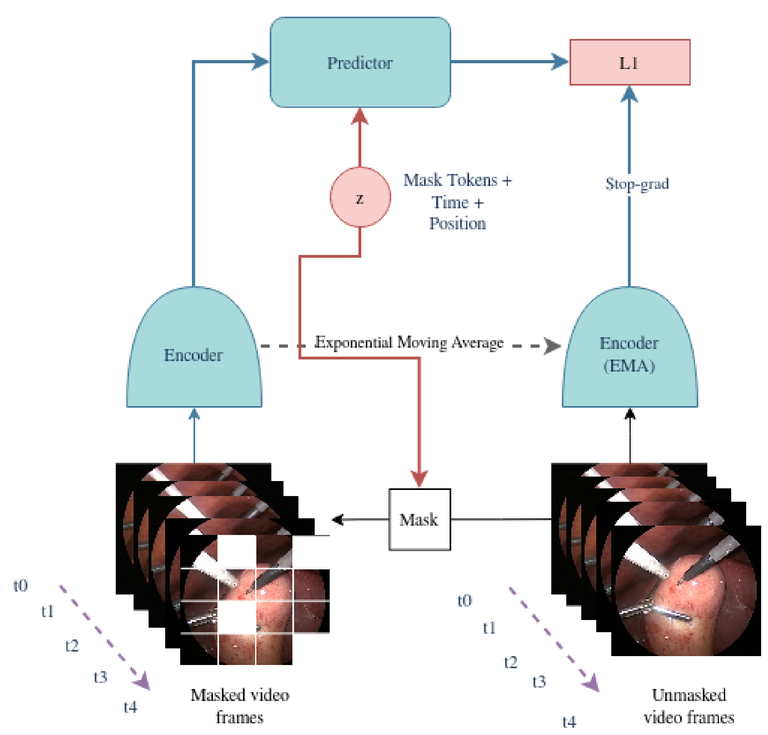

Figure 3. V-JEPA Architecture

Representation Analysis

To understand where and how the model learns, this project employs:

- Layer Probing: Identifying "Emergence Zones" where phase semantics become linearly separable.
- UMAP Visualization: Projecting high-dimensional embeddings into 2D/3D to assess cluster stability and phase transitions.

Dataset

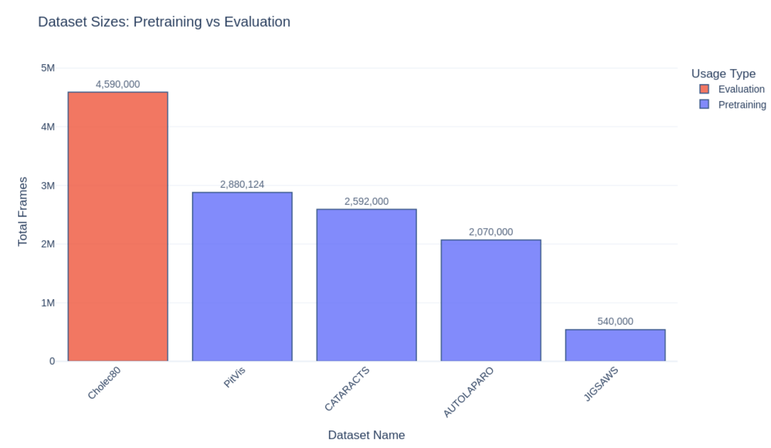

Figure 4. Dataset distribution [8] [4] [1] [9] [5]

References

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